From 500 to a General Formula: Exploring a Floor Identity

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1 THE PROBLEM AND A CHANGE OF PERSPECTIVE

The original problem. Prove that there exists a real number w such that

$$\boxed{\sum_{k=0}^{500} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor = \lfloor x \rfloor - \lfloor wx \rfloor \qquad \text{for every } x \in \mathbb{R}.} \tag{1}$$

Before trying to solve the problem, let us test a special value of x. Taking $x = 2^{501}$, we have

$$\left\lfloor \frac{2^{501}+2^k}{2^{k+1}} \right\rfloor = 2^{500-k}, \qquad 0 \leqslant k \leqslant 500.$$

Therefore,

$$\sum_{k=0}^{500} \left\lfloor \frac{2^{501}+2^k}{2^{k+1}} \right\rfloor = \sum_{k=0}^{500} 2^{500-k} = 2^{501} - 1.$$

If the identity holds for every real x, we must have

$$2^{501} - 1 = 2^{501} - \left\lfloor 2^{501} w \right\rfloor,$$

and hence

$$\left\lfloor 2^{501} w \right\rfloor = 1.$$

By the definition of the floor function,

$$1 \leqslant 2^{501} w < 2.$$

Dividing by the positive number 2^{501} gives

$$\boxed{2^{-501} \leqslant w < 2^{-500}.}$$

This gives an interval for w, but can we narrow it down? Let us generalize our choice of x by taking

$$x = 2^{500+s}, \qquad s \in \mathbb{Z}_{\geqslant 1}.$$

For every $0 \leqslant k \leqslant 500$, we have

$$\left\lfloor \frac{2^{500+s}+2^k}{2^{k+1}} \right\rfloor = 2^{499+s-k}.$$

Consequently,

$$\sum_{k=0}^{500}\left\lfloor\frac{2^{500+s}+2^k}{2^{k+1}}\right\rfloor=\sum_{k=0}^{500}2^{499+s-k}$$
$$=2^{500+s}-2^{s-1}.$$

If the original identity holds for every real x, then

$$\left\lfloor 2^{500+s}w\right\rfloor=2^{s-1}.$$

By the definition of the floor function,

$$2^{s-1}\leqslant 2^{500+s}w<2^{s-1}+1.$$

Dividing by the positive number 2^{500+s}, we obtain

$$\boxed{2^{-501}\leqslant w<2^{-501}+2^{-(500+s)}.}$$

This inequality holds for every positive integer s. Letting $s\to\infty$, we conclude that

$$\boxed{w=2^{-501}.}$$

This is a necessary condition, not yet a proof that this value works for all real x.

But is there anything mathematically special about 500? The summands follow the same pattern, regardless of where we stop. This suggests that 500 may not be essential to the problem.

Instead of trying to handle all 501 summands directly, let us replace 500 by a nonnegative integer m. Since the unknown number may depend on m, we denote it by w_m.

We are thus led to a more general problem: for each $m\in\mathbb{Z}_{\geqslant 0}$, find a real number w_m such that

$$\sum_{k=0}^{m}\left\lfloor\frac{x+2^k}{2^{k+1}}\right\rfloor=\lfloor x\rfloor-\lfloor w_m x\rfloor\qquad\text{for every } x\in\mathbb{R}.\tag{2}$$

At first sight, replacing 500 by an arbitrary m may seem to make the problem more difficult. However, this generalization allows us to investigate the simplest cases, starting with $m=0$ and then $m=1$.

Since the identity is required to hold for *every* real x, we can choose special values of x that simplify the floor expressions. These choices may help us discover a candidate for w_m, which we must then verify for all real x.

2 EXPLORING THE GENERALIZED PROBLEM

2.1 The case $m=0$

Equation (2) becomes

$$\left\lfloor\frac{x+1}{2}\right\rfloor=\lfloor x\rfloor-\lfloor w_0x\rfloor\,.\tag{3}$$

Step 1: Find a candidate.

Take $x = 2n-1$ with $n \in \mathbb{Z}$. For the argument below, we choose $n \geqslant 1$, so that $2n-1 > 0$. Then

$$\left\lfloor \frac{(2n-1)+1}{2} \right\rfloor = n, \qquad \lfloor 2n-1 \rfloor = 2n-1.$$

If (3) holds for all x, it follows that

$$n = 2n-1-\lfloor w_0(2n-1) \rfloor,$$

and hence

$$\lfloor w_0(2n-1) \rfloor = n-1.$$

By the definition of the floor function,

$$n-1 \leqslant w_0(2n-1) < n.$$

Our choice $n \geqslant 1$ ensures that $2n-1 > 0$. Dividing by this positive number gives

$$\frac{n-1}{2n-1} \leqslant w_0 < \frac{n}{2n-1}.$$

Since the assumed identity holds for every real x, these inequalities hold for every positive integer n. Both bounds tend to $\frac{1}{2}$ as $n \to \infty$. Therefore,

$$\boxed{w_0 = \frac{1}{2}.}$$

This is a *necessary condition*. So far, we have not proved that this value works for every real x.

Step 2: Verify the candidate.

We recall the following special case of *Hermite's floor identity*:

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor, \qquad y \in \mathbb{R}. \tag{4}$$

A proof is given in the appendix.

Taking $y = x/2$, we obtain

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

Hence (3) holds for every real x with $w_0 = \frac{1}{2}$.

2.2 The case $m = 1$

Now the desired identity becomes

$$\left\lfloor \frac{x+1}{2} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \lfloor x \rfloor - \lfloor w_1 x \rfloor. \tag{5}$$

We follow the same two-step strategy.

Step 1: Find a candidate.

As in the first case, take $x = 2n-1$ with $n \in \mathbb{Z}$.

To simplify the second floor term as well, choose $n = 2s$ with $s \in \mathbb{Z}_{>0}$. Thus our test values are

$$x = 4s-1 > 0.$$

Then

$$\left\lfloor \frac{x+1}{2} \right\rfloor = 2s, \qquad \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor s + \frac{1}{4} \right\rfloor = s.$$

Thus (5) implies

$$3s = 4s - 1 - \lfloor (4s-1)w_1 \rfloor,$$

or equivalently,

$$\lfloor (4s-1)w_1 \rfloor = s - 1.$$

By the definition of the floor function,

$$s - 1 \leqslant (4s-1)w_1 < s.$$

Our choice $s \geqslant 1$ ensures $4s - 1 > 0$. Dividing by this positive number gives

$$\frac{s-1}{4s-1} \leqslant w_1 < \frac{s}{4s-1}.$$

Since the assumed identity holds for every real x, these inequalities hold for every $s \in \mathbb{Z}_{>0}$.

Letting $s \to \infty$, we obtain the necessary value

$$\boxed{w_1 = \frac{1}{4}.}$$

Step 2: Verify the candidate.

Apply (4) twice. With $y = x/4$,

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor = \left\lfloor \frac{x}{2} \right\rfloor. \tag{6}$$

With $y = x/2$,

$$\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor. \tag{7}$$

Adding these equalities and cancelling the common term $\lfloor x/2 \rfloor$ gives

$$\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor.$$

This proves (5) for every $x \in \mathbb{R}$ with $w_1 = \frac{1}{4}$.

What have we discovered?

The first two cases give

$$w_0 = \frac{1}{2}, \qquad w_1 = \frac{1}{4},$$

suggesting that

$$w_m = \frac{1}{2^{m+1}}$$

might work in general.

More importantly, the proof for $m = 1$ reveals a structure: applying Hermite's identity at successive scales allows intermediate terms to cancel. This suggests a way to prove the conjecture for arbitrary m.

2.3 The general case

We now prove that the proposed formula works for every nonnegative integer m.

Theorem 2.1. *For every $m \in \mathbb{Z}_{\geqslant 0}$ and every $x \in \mathbb{R}$,*

$$\boxed{\sum_{k=0}^{m} \left\lfloor \frac{x+2^k}{2^{k+1}} \right\rfloor + \left\lfloor \frac{x}{2^{m+1}} \right\rfloor = \lfloor x \rfloor .} \tag{8}$$

In particular, $w_m = 2^{-(m+1)}$ satisfies (2).

Proof. The case $m = 0$ has already been proved. We may therefore assume $m \geqslant 1$.

The key is again Hermite's identity

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .$$

Use it successively with

$$y = \frac{x}{2^{m+1}},\ \frac{x}{2^m},\ \dots,\ \frac{x}{4},\ \frac{x}{2}.$$

This gives

$$\begin{aligned}
\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor &= \left\lfloor \frac{x}{2^m} \right\rfloor, \\
\left\lfloor \frac{x}{2^m} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor &= \left\lfloor \frac{x}{2^{m-1}} \right\rfloor, \\
&\vdots \\
\left\lfloor \frac{x}{4} \right\rfloor + \left\lfloor \frac{x+2}{4} \right\rfloor &= \left\lfloor \frac{x}{2} \right\rfloor, \\
\left\lfloor \frac{x}{2} \right\rfloor + \left\lfloor \frac{x+1}{2} \right\rfloor &= \lfloor x \rfloor .
\end{aligned}$$

Add all these equations. Every intermediate term $\left\lfloor x/2^j \right\rfloor$, $1 \leqslant j \leqslant m$, appears once on each side and cancels. What remains is

$$\left\lfloor \frac{x}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^m}{2^{m+1}} \right\rfloor + \left\lfloor \frac{x+2^{m-1}}{2^m} \right\rfloor + \cdots + \left\lfloor \frac{x+1}{2} \right\rfloor = \lfloor x \rfloor ,$$

which is exactly (8). □

Returning to the original problem, we take $m = 500$. Thus the required identity holds for every $x \in \mathbb{R}$ with

$$\boxed{w = \frac{1}{2^{501}}.}$$

3 REFLECTION: FROM A SPECIFIC PROBLEM TO A GENERAL PROOF

Our exploration involved two different kinds of generalization. First, while keeping the original upper limit 500, we replaced the test value $x = 2^{501}$ by a family of values $x = 2^{500+s}$. The resulting inequalities became increasingly precise as s grew, allowing us to determine the only possible value of w. However, this necessary condition did not establish the identity for every real x.

We then questioned whether the number 500 was essential. Replacing it by a nonnegative integer m turned a single problem into a family of related problems. At first, this might seem to make the problem more difficult. In fact, it allowed us to investigate the simplest cases, $m = 0$ and $m = 1$, which were not available in the original formulation.

For each of these cases, we chose special values of x to discover a candidate for w_m, and then used Hermite's floor identity to verify it for every real x. The cases suggested the formula

$$w_m = \frac{1}{2^{m+1}},$$

while the repeated cancellation in the proofs revealed how to establish it for arbitrary m.

These steps illustrate the distinction between *discovering* a possible result and *proving* it. Special test values can impose necessary conditions; they do not, by themselves, prove an identity for all real numbers. To complete the argument, we needed a proof of the proposed formula in full generality.

Sometimes, generalizing a problem makes it easier to understand: one generalization can sharpen a candidate, while another can reveal the structure needed for its proof.

A A PROOF OF HERMITE'S FLOOR IDENTITY

We prove that, for every $y \in \mathbb{R}$,

$$\boxed{\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = \lfloor 2y \rfloor .}$$

Proof. Let $n = \lfloor y \rfloor$. Then $y = n + r$ for some $0 \leqslant r < 1$. Hence

$$\lfloor y \rfloor + \left\lfloor y + \frac{1}{2} \right\rfloor = 2n + \left\lfloor r + \frac{1}{2} \right\rfloor, \qquad \lfloor 2y \rfloor = 2n + \lfloor 2r \rfloor .$$

It remains to show that $\left\lfloor r + \frac{1}{2} \right\rfloor = \lfloor 2r \rfloor$.

If $0 \leqslant r < \frac{1}{2}$, both sides equal 0. If $\frac{1}{2} \leqslant r < 1$, both sides equal 1. Thus the identity holds for every $y \in \mathbb{R}$. □